

Adaptivity and Optimism

An Improved Exponentiated Gradient Algorithm

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1.1 Hedge Algorithm: Protocol and Bound

Algorithm: Hedge — expert advice protocol

- 1: Initialize weights $w_{1,i} = 1$ for all i .
 - 2: For $t = 1, \dots, T$, form $p_{t,i} = \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$ learner plays p_t .
 - 3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.
 - 4: For each i , update $w_{t+1,i} = w_{t,i} \exp(-\eta z_{t,i})$.
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Regret: for $u \in \Delta_n$, $R_{T(u)} := \sum_{t=1}^T (p_t - u)^\top z_t$.

Standard upper bound.

$$R_{T(u)} \leq \frac{\log n}{\eta} + \eta \sum_{t=1}^T \|z_t\|_\infty^2$$

1.2 Adaptive Weight Update: MW2

Algorithm: MW2 — adaptive weights

- 1: Initialize weights $w_{1,i} = 1$ for all i .
 - 2: For $t = 1, \dots, T$, form $p_{t,i} = \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$ learner plays p_t .
 - 3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.
 - 4: For each i , update $w_{t+1,i} = w_{t,i} (1 - \eta z_{t,i})$.
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Bound:

$$\text{Regret}(u) \leq \frac{\log n}{\eta} + \eta \sum_i u_i \sum_t z_{t,i}^2$$

The second term is now averaged by the comparator u .

1.3 Why MW2 Improves the Bound

Hedge pays for the worst coordinate on every round:

$$\eta \sum_t \|z_t\|_\infty^2$$

MW2 pays only for the comparator-weighted coordinates:

$$\eta \sum_i u_i \sum_t z_{t,i}^2$$

For the best expert $u = e_{i^*}$, this becomes

$$\eta \sum_t z_{t,i^*}^2$$

Irrelevant noisy experts no longer enlarge the second term.

1.4 Optimistic Learning: Hints

Algorithm: Optimistic EG – hints

- 1: Initialize scores $\theta_{1,i} = 0$ for all i .
 - 2: For $t = 1, \dots, T$, using hint m_t , form $p_{t,i} \propto \exp(\theta_{t,i} - \eta m_{t,i})$ learner plays p_t .
 - 3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.
 - 4: For each i , update $\theta_{t+1,i} = \theta_{t,i} - \eta z_{t,i}$.
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Bound:

$$R_{T(u)} \leq \frac{\log n}{\eta} + \eta \sum_t \|z_t - m_t\|_\infty^2$$

The second term now measures hint error. With $m_t = z_{t-1}$, this becomes a path-length quantity.

1.5 Two Ways to Improve the Second Term

We now have two independent changes to the Hedge template:

Adaptive weights

Change the weight update.

Worst-coordinate scale becomes comparator scale:

$$\sum_i u_i \sum_t z_{t,i}^2.$$

Optimism / hints

Change the probability before observing z_t .

Loss scale becomes hint-error scale:

$$\sum_t \|z_t - m_t\|_\infty^2.$$

The paper combines both changes in one algorithm.

1.6 Why Improve the Second Term?

Many refinements keep the first term $\frac{\log n}{\eta}$ and replace the second term by a more local quantity.

- **Standard EG:** $\sum_{t=1}^T \|z_t\|_{\infty}^2$.
- **Adaptive / MW2:** $\sum_{i=1}^n u_i \sum_{t=1}^T z_{t,i}^2$.
- **Optimistic methods:** $\sum_{t=1}^T \|z_t - m_t\|_{\infty}^2$.
- **This paper:** $\sum_{i=1}^n u_i \sum_{t=1}^T (z_{t,i} - m_{t,i})^2$.

Drawback of $\|z_t\|_{\infty}^2$: it uses the worst coordinate at each round, not the comparator chosen in hindsight.

A noisy irrelevant expert can enlarge the bound even when the best expert is stable or predictable.

2.1 Algorithm: Adaptive Optimistic EG

Algorithm: Adaptive Optimistic EG — adaptive weights + hints

1: Initialize weights $w_{1,i} = 1$ for all i .

For $t = 1, \dots, T$, using hint m_t , form

2:
$$p_{t,i} = \frac{w_{t,i} \exp(-\eta m_{t,i})}{\sum_{j=1}^n w_{t,j} \exp(-\eta m_{t,j})}$$

learner plays p_t .

3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.

4: For each i , update $w_{t+1,i} = w_{t,i} \exp\left(-\eta z_{t,i} - \eta^2 (z_{t,i} - m_{t,i})^2\right)$.

The red terms use the hint in prediction and penalize inaccurate hints in the update.

2.2 Upper Bound

Theorem (Adaptive Exponentiated Gradient)

If $\|z_t\|_\infty \leq 1$, $\|m_t\|_\infty \leq 1$, and $0 < \eta \leq \frac{1}{4}$, then for every $u \in \Delta_n$,

$$\text{Regret}(u) \leq \frac{\log n}{\eta} + \eta \sum_i u_i \sum_t (z_{t,i} - m_{t,i})^2.$$

This is exactly the desired combination: the comparator chooses the relevant coordinates, and the hint error chooses the relevant scale.

The step-size condition is for the proof inequality. Since $|z_{t,i} - m_{t,i}| \leq 2$, $\eta \leq \frac{1}{4}$ gives $|\eta(z_{t,i} - m_{t,i})| \leq \frac{1}{2}$, so $\exp(-x - x^2) \leq 1 - x$ applies.

3.1 Hint Choices: Path-Length and Variance

The second term is the hint error $H_{i(m)} := \sum_t (z_{t,i} - m_{t,i})^2$. Different hints make $H_{i(m)}$ small under different patterns.

Last-loss hint

$$m_t = z_{t-1}$$

Controls: path length of expert i :

$$D_i := \sum_t (z_{t,i} - z_{t-1,i})^2.$$

Here $H_{i(m)} = D_i$, so

$$\text{Regret}(i^*) \leq \frac{\log n}{\eta} + \eta D_{i^*}.$$

Interpretation: good when the best expert changes slowly from one round to the next.

Past-average hint

$$m_t = \frac{1}{t} \sum_{s=1}^{t-1} z_s$$

Controls: variance around a stable mean:

$$V_i := \sum_t (z_{t,i} - \mu_i)^2, \quad \mu_i := \frac{1}{T} \sum_t z_{t,i}.$$

Paper's lemma:

$$\sum_t (z_{t,i} - m_{t,i})^2 \leq 2V_i + 6.$$

$$\text{Regret}(i^*) \leq \frac{\log n}{\eta} + \eta(2V_{i^*} + 6).$$

Good when: the best expert's losses stay near one fixed average, even with noise.

4.1 Proof Tool: Fenchel Conjugate

The paper writes EG through the Fenchel conjugate of negative entropy:

$$\psi(w) = \sum_i w_i \log w_i, \quad \psi^*(\beta) = \log \left(\sum_i \exp(\beta_i) \right).$$

The conjugate turns scores into probabilities:

$$\nabla \psi^*(\beta)_i = \frac{\exp(\beta_i)}{\sum_j \exp(\beta_j)}.$$

So the algorithm can be written as

$$p_t = \nabla \psi^*(\beta_t - \eta m_t),$$

and the proof tracks how ψ^* changes after each update.

4.2 Matrix Extension

Replace distributions by density matrices and losses by symmetric matrices:

$$W_t, U \in \mathcal{S}_+^n, \quad \text{tr}(W_t) = \text{tr}(U) = 1, \quad \ell_{t(W_t)} = \text{tr}(W_t Z_t), \quad \|Z_t\|_{\text{op}} \leq 1.$$

Matrix regret is measured against a fixed density matrix U :

$$\text{Regret}(U) := \sum_t \text{tr}((W_t - U)Z_t).$$

General hint version (Proposition 4.1):

$$\text{Regret}(U) \leq \frac{\log n}{\eta} + \eta \sum_t \text{tr}(U(Z_t - M_t)^2).$$

Path-length hint: with $M_t = Z_{t-1}$,

$$\text{Regret}(U) \leq \frac{\log n}{\eta} + \eta \sum_t \text{tr}(U(Z_t - Z_{t-1})^2).$$